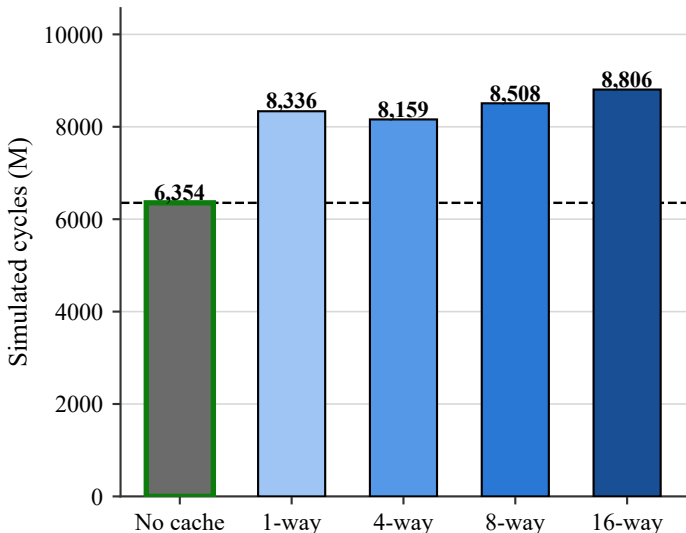


**2000-read workload, 50MB DB: cache is worse
than no cache at every width tested**



35.1M bases total (2000 reads), sample_targeted (50MB) DB. Dashed line: no-cache (S0) cycle count - every cache width sits above it, worsening monotonically past 4-way. Width shown = the SOFTWARE S2 lookup-cache's associativity. Real Luna HARDWARE cache is fixed throughout: L1d 48KB/12-way, L1i 32KB/8-way, L2 2MB/16-way, L3 105MB/15-way.